Aramark Procurement Analytics — Full Project Report

Course: CS 562 — Big Data **Dataset:** Aramark 2025 US Procurement Data (~43 million rows, ~1 billion cells) **Source file:** `gs://cs-562-aramark-project/Andrew_Meszaros_SRF_2026-04-01-0936.csv` **Tooling:** Google Cloud Storage, BigQuery, Google Colab, Python (pandas, scikit-learn, matplotlib, seaborn), NOAA climdiv

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Introduction

This report walks through an analytics project on a 43-million-row procurement dataset from Aramark, covering every US transaction in calendar year 2025. The work is organized into seven analytical phases, each building on the last: profiling the raw data, moving it into a cloud warehouse, computing aggregations, quantifying supply-chain risk, pulling in external weather data, testing weather-driven patterns, and finding hidden customer archetypes via machine learning.

The document is structured so that every technical term, formula, and design decision is explained at the point it first appears. By the end, a reader with no prior exposure to the project should be able to walk through every chart, describe what it shows, and defend the methodology behind it. A short executive summary at the end consolidates the headline findings in one place.

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1. Dataset Description

The source CSV contains 23 columns. After an initial data quality review, eleven columns were retained as the core analytical set for in-memory operations, and additional columns were pulled back in selectively once the data moved to BigQuery (where column count no longer affects query cost meaningfully).

1.1 Columns used in the analysis

Column	Type	Role
year_month	Integer (YYYYMM)	Monthly time grain
business_entity_type	String	GPO vs Managed Services split
customer_market_segment_id	String	Industry segment identifier
state, city	String	Geographic rollups
number_of_rooms	Integer	Operational scale (hospitality signal)
distributor_group	String	Primary supplier grouping
category_level_1 / level_2 / level_3 / level_4	String	Product hierarchy
manufacturer_id	String	Upstream supplier identity
customer_id	String	DISTINCT counts for segmentation
spend_random_factor	Float	USD spend proxy (randomized)

1.2 Columns dropped and the reasons

- `year_name` — constant (2025)
- `country` — constant (US)
- `client_id`, `zip` — high-cardinality identifiers with no aggregation value or redundant with state/city
- `customer_brand_id` — approximately 21% missing
- `customer_brand_parent_id` — approximately 73% missing (too sparse)
- `ecommerce_status` — approximately 88% missing (unusable)
- `distributor_id` — redundant with `distributor_group`
- `category_id` — redundant with the explicit category hierarchy

1.3 The Spend Random Factor caveat

The dollar values in this column are **randomized** for confidentiality reasons. Directional and relative claims remain valid: if one state's spend proxy is twice another's, the underlying real spend is also approximately twice as large. Ratios, shares, and rankings are trustworthy. Absolute dollar statements (e.g. "FOOD spend was $3.7 billion") should be read as spend-proxy language, not as audited financial totals.

1.4 Scope limitations we declared upfront

- **No forecasting.** The dataset covers exactly 12 months of one calendar year. Any credible forecasting model needs multi-year data to separate seasonal patterns from trend.

- **No causal claims.** All relationships described here are correlations or structural observations. With 12 time points per state, causal identification is not defensible.
- **No absolute dollar claims.** See the `Spend Random Factor` caveat above.

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2. Architecture and Tooling

2.1 System in words

1. **Source layer.** The raw CSV sits in Google Cloud Storage, unchanged.
2. **Warehouse layer.** A one-time load operation ingests the CSV into a typed BigQuery table (`aramark_spend.raw_spend`, 43M rows, 9.4 GB). BigQuery is Google Cloud's serverless, columnar, distributed SQL warehouse.
3. **Aggregation layer.** 16 BigQuery views materialize common rollups (by state, by category, by distributor, and cross-cuts). A **view** is a stored SQL query that behaves like a table: every query against it re-runs the underlying SQL against the source table, so views cost no storage and cannot drift from the source data.
4. **Analysis layer.** Colab notebooks issue SQL against the views, pull small result sets into pandas for visualization, and fit machine learning models on stratified samples where needed.
5. **External data layer.** NOAA climdiv climate data is downloaded, parsed, and loaded as a second BigQuery table (`weather_state_monthly`). It joins cleanly to the spend aggregations on (state, year_month).

2.2 Why BigQuery was the right call

The initial approach in Phase 0 used pandas chunking — streaming the CSV in 100,000-row chunks and accumulating aggregations in Python dictionaries. That worked but took roughly 10 minutes per full aggregation and had to be re-run after every Colab disconnect. For a project that required 10+ aggregations plus joins with external data, pandas chunking would have taken several hours per iteration.

Migrating the data into BigQuery once, and then issuing SQL queries against the persistent table, reduced most aggregations to 2–5 seconds. Views enable a clean separation between the raw data, the reusable aggregations, and the analysis that consumes them. This is the same pattern used by enterprise analytics teams, not a course-project workaround.

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3. Phase 0 – 2: From Raw CSV to BigQuery Warehouse

3.1 Phase 0 — Access and profiling

Initial access used Python's `gcsfs` library to open the CSV directly from GCS. A 5-row preview was pulled to confirm the column structure, then a 100,000-row chunk was profiled to compute missing-value counts and data types.

This profiling surfaced the missing-value problems (ecommerce status, customer brand parent, manufacturer ID)

and the redundant identifiers (zip vs state/city, distributor ID vs distributor group) that drove the column selection decisions above.

A chunk-based aggregation over the full 427-chunk dataset produced the first real finding: `MASTER DISTRIBUTOR` accounts for roughly 3.28 billion in spend proxy and 28.4 million transactions — already signaling heavy concentration before any formal metric was computed.

3.2 Phase 1 — Loading into BigQuery

We defined an explicit BigQuery schema with snake_case column names and correct types (`INTEGER` for year_month and number_of_rooms, `FLOAT64` for spend, `STRING` for everything else including zip to preserve leading zeros). Explicit schema is preferred over auto-detection because the column names are clean and the types are deterministic.

The load job invokes `client.load_table_from_uri(...)` with `source_format=CSV`, `skip_leading_rows=1`, and `write_disposition="WRITE_TRUNCATE"`. BigQuery parallelizes the read directly from GCS — nothing transits through Colab. The full 43M-row load completed in roughly 3 minutes and produced a 9.4 GB table.

A sanity check re-ran the Phase 0 distributor aggregation in pure SQL against the new BigQuery table. The result matched the pandas chunking output exactly, confirming that the load was complete and the types were correct.

3.3 Phase 2 — Core and cross-dimensional aggregations, materialized as views

With the data in BigQuery, every aggregation becomes a SQL query. We computed eight single-dimension aggregations and six cross-dimensional ones:

Single-dimension:

- Distributor group (national)
- Category level 1
- Category level 1 × 2 (nested)
- State
- Year-month (monthly trend)
- Business entity type (GPO vs Managed Services)
- Market segment
- Number-of-rooms bands (0, 1–100, 101–300, 301–500, 500+)

Cross-dimensional:

- Category L1 × state
- Category L1 × year-month
- Distributor × state (full rollup, not filtered to top-1)
- Category L1 × entity type (with percentage within entity)
- Distributor × room band
- Category L1 → L2 → L3 → L4 full drill-down

Each aggregation was wrapped in a `CREATE OR REPLACE VIEW` statement. Fourteen views total at the end of

Phase 2, plus two more added in Phase 3. The views follow a consistent naming convention: `agg_<dimension>` for single-dim, `agg_<dim1>_<dim2>` for cross-cuts.

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4. Phase 3: Concentration and Risk Analysis

Phase 3 is where the project starts quantifying risk using real metrics. This section introduces five concepts that appear throughout the rest of the project, so every term is defined here with its formula and interpretation.

4.1 Why concentration matters

Phase 2 showed that `MASTER DISTRIBUTOR` carries roughly 53% of all spend. That is a headline, but it is not yet a risk metric. Phase 3 converts raw concentration into quantified risk using metrics that the US Department of Justice (DOJ) and Federal Trade Commission use in antitrust reviews. The output is a state-level risk scorecard that every downstream phase can query.

4.2 Concept — the Lorenz curve

What it is. The Lorenz curve is a simple inequality visualization. Sort all distributor groups from smallest to largest by spend. Plot the cumulative share of groups on the x-axis and the cumulative share of spend on the y-axis. If every group had equal spend, the curve would be the 45-degree diagonal line. The more the curve bows below the diagonal, the more unequally the spend is distributed.

How to read it. The bigger the area between the diagonal and the Lorenz curve, the more unequal the distribution. In a perfectly equal world the area is zero. In a one-firm monopoly the area approaches a triangle.

4.3 Concept — the Gini coefficient

What it is. A single number from 0 to 1 that summarizes the Lorenz curve:

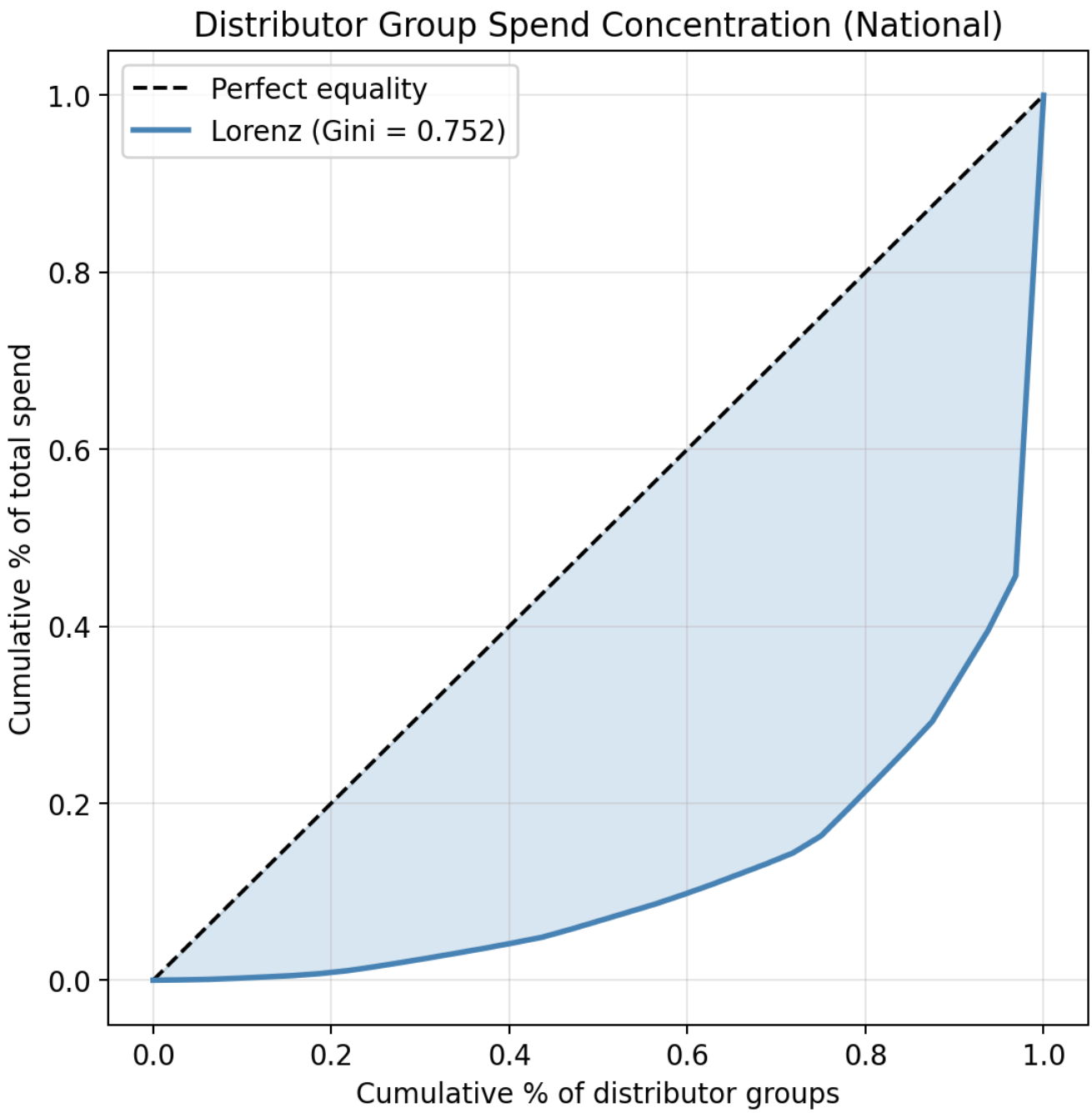
$G = 1 - 2 \times$ (area under the Lorenz curve)

The area under a perfectly equal Lorenz curve (the diagonal) is 0.5, so $G = 0$. The area under a monopoly Lorenz curve approaches 0, so G approaches 1.

Rules of thumb:

- 0.0 – 0.3 — relatively equal
- 0.3 – 0.5 — moderate inequality
- 0.5 – 0.8 — high inequality
- above 0.8 — extreme concentration

The code. We pull the full list of distributor groups sorted ascending by total spend. `np.cumsum(spend) / spend.sum()` gives the cumulative share of spend; `np.arange(1, n+1) / n` gives the cumulative share of groups. The Gini integral is computed with `np.trapz(cum_share, cum_pct)` — numpy's trapezoidal integration, which approximates the area under a curve by stacking trapezoids between adjacent data points.



What this chart shows. The Lorenz curve is pushed hard toward the bottom-right corner. The bottom 70% of distributor groups together account for only a sliver of total spend. The Gini coefficient quantifies this as extreme concentration. This is structural: the Aramark procurement network has a small number of dominant channels and a long tail of tiny specialty distributors.

4.4 Concept — top-N share

What it is. The simplest concentration metric: what percentage of total spend flows through the top N distributor groups? Compute the total spend, sum the spend of the top N, divide.

Results from the national-level SQL aggregation:

- Top 1: 54.30%
- Top 3: 65.63%
- Top 5: 74.12%

- Top 10: 86.89%

Translation: the remaining 23 distributor groups combined account for approximately 13% of spend. `MASTER DISTRIBUTOR` alone handles more spend than all non-top-10 distributors combined, by a factor of more than four.

4.5 Concept — the Herfindahl-Hirschman Index (HHI)

What it is. HHI is the sum of squared market shares, expressed as percentages:

$HHI = \Sigma (share_i)^2$

A monopoly (one firm with 100% share) produces HHI = 10,000. Perfect competition with many tiny firms produces HHI near zero.

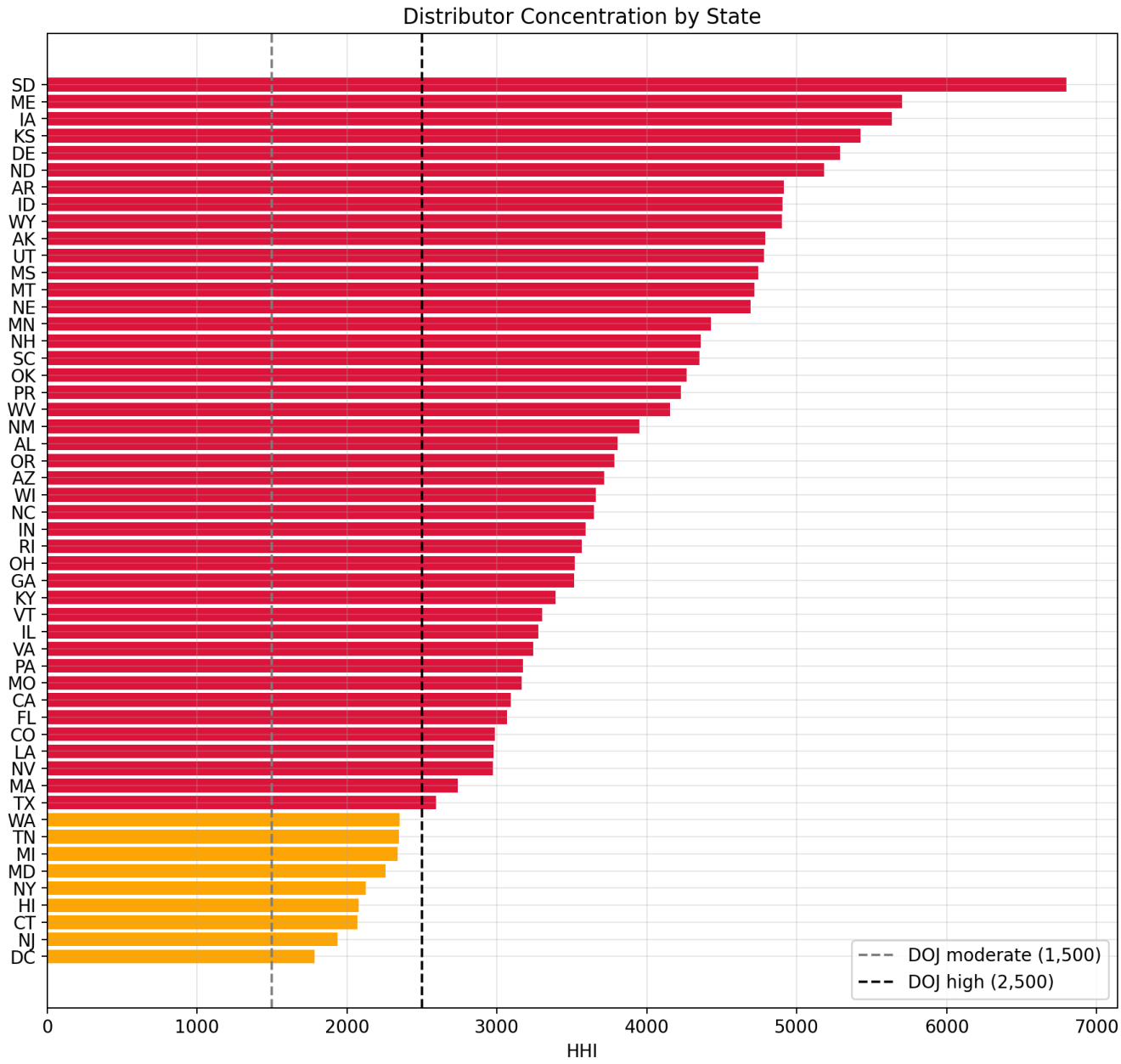
Concept — DOJ concentration thresholds. The US Department of Justice uses specific HHI thresholds for antitrust review of mergers. These are widely-cited, externally-validated cutoffs that give us a defensible scale for describing concentration:

- Below 1,500 — **competitive market**
- 1,500 to 2,500 — **moderate concentration**
- Above 2,500 — **highly concentrated** (flag for regulatory scrutiny)

We are not doing an antitrust review, but applying the DOJ scale to state-level distributor concentration gives us a meaningful and defensible risk tier system. A state with HHI above 2,500 has a distributor landscape that, in a merger context, would trigger regulatory attention. That is the bar we use for "HIGH risk."

The SQL implementation. Computing HHI in SQL requires each distributor's share within each state, squared, and summed per state. The challenge is computing the within-state share in one pass. We use a **window function**: `100.0 * total_spend / SUM(total_spend) OVER (PARTITION BY state)` computes each distributor's share of its state's total spend in the same pass. Then `SUM(POW(share_pct, 2))` per state gives the HHI. A second window function `ROW_NUMBER() OVER (PARTITION BY state ORDER BY total_spend DESC)` is used in the same query to extract the top-1, top-3, and top-5 shares per state.

Data quality filter. The state column has three anomalous values that we exclude from Phase 3: `GEORGIA` (full name, only 2 rows — a data entry error for GA), `AB` (one row, a Canadian province violating the US-only scope), and `PR` (Puerto Rico, only 2,082 rows — kept but flagged in the scorecard).



What this chart shows. Every single state exceeds the moderate concentration threshold. 42 states exceed the high-concentration threshold. The most concentrated states are small-spend states (SD at HHI 6,798, ME at 5,701, IA at 5,634) where fewer distributor relationships exist and `MASTER DISTRIBUTOR` fills the gap almost unilaterally. The least concentrated states (DC at 1,786, NJ at 1,936, CT at 2,069) are large urban markets where more distribution alternatives naturally exist. Every state's top-1 distributor is `MASTER DISTRIBUTOR` — without exception.

4.6 Category-level concentration

State-level HHI tells us where concentration is high. Category-level HHI tells us in what product line. We created a new BigQuery view, `agg_distributor_category`, aggregating spend by (category_level_1, distributor_group), and applied the same HHI math.

The interpretation has a structural-vs-operational distinction:

- A high-HHI category whose top distributor is a **specialty channel** (for example, MEDICAL dominated by a distributor literally named MEDICAL, GOLF EQUIPMENT dominated by GOLF EQUIPMENT & PARTS) is **structurally concentrated**. Specialty products naturally flow through

narrow channels. This is not operational risk.

- A high-HHI category whose top distributor is `MASTER DISTRIBUTOR` is concentration that Aramark created through channel strategy, not through product structure. This is where operational risk actually sits.

Observed patterns. FOOD — the largest category at roughly $3.7B spend proxy — has HHI of 5,438 with `MASTER DISTRIBUTOR` at 72.85%. This is the single highest-impact concentration point in the entire dataset. DISPOSABLES, CHEMICALS AND CLEANING, and BEVERAGE all route through `MASTER DISTRIBUTOR` as well, confirming the central-logistics-channel pattern. MEDICAL, GOLF, POOL, and CLOTHING show the structural pattern (a specialty distributor at the top).

4.7 Manufacturer-level concentration (upstream)

`MASTER DISTRIBUTOR` is a logistics pass-through. It does not manufacture ground beef or paper towels; it moves them. Real upstream concentration sits at the `manufacturer_id` level. We aggregated (state, manufacturer_id) spend and computed HHI for each state, with `manufacturer_id IS NOT NULL` filtering.

Documented caveat. `manufacturer_id` is approximately 34% missing in the raw data. The manufacturer HHI therefore reflects the known portion of the manufacturer mix only. Directional comparisons across states remain valid; absolute values should be read with the coverage caveat in mind.

Key finding. Manufacturer HHI is almost uniformly low — most states below 250. The single outlier is Puerto Rico at HHI 5,140, which reflects PR's tiny sample of only 4 manufacturers rather than any real upstream concentration. Top-1 manufacturer shares nationwide rarely exceed 10%.

This is the reassuring finding of the project. The risk concentration sits in the logistics layer, not the source layer. If `MASTER DISTRIBUTOR` were disrupted, the products would still exist in the market — alternative distributors would be able to source the same goods. The risk is continuity of operations, not continuity of supply.

4.8 The unified state risk scorecard

The final deliverable of Phase 3 is `agg_state_risk_scorecard`, a BigQuery view that assembles per-state distributor HHI, manufacturer HHI, top-1 and top-3 shares, top-distributor name, total state spend, and a three-level risk flag. Every downstream phase queries this single view to tell the state-risk story.

Risk flag logic:

- **HIGH** — distributor HHI > 2,500, OR manufacturer HHI > 2,500, OR top-1 distributor share > 60%
- **MODERATE** — distributor HHI between 1,500 and 2,500
- **LOW** — otherwise

Results:

- 42 states HIGH
- 10 states MODERATE
- 0 states LOW

Every MODERATE state triggers on distributor HHI being between 1,500 and 2,500. Every HIGH state triggers predominantly on `dist_hhi > 2,500` or `top1_share > 60%`. Only Puerto Rico triggers via manufacturer HHI.

Top 15 States by Distributor HHI — Risk Scorecard

State	Spend	Top Distributor	Top-1 Share	Dist HHI	Mfr HHI	Risk
SD	$9M	MASTER DISTRIBUTOR	82.2%	6,798	210	HIGH
ME	$13M	MASTER DISTRIBUTOR	75.0%	5,701	137	HIGH
IA	$58M	MASTER DISTRIBUTOR	74.5%	5,634	108	HIGH
KS	$29M	MASTER DISTRIBUTOR	73.2%	5,425	137	HIGH
DE	$20M	MASTER DISTRIBUTOR	72.2%	5,287	211	HIGH
ND	$9M	MASTER DISTRIBUTOR	70.9%	5,181	233	HIGH
AR	$39M	MASTER DISTRIBUTOR	69.5%	4,914	177	HIGH
ID	$24M	MASTER DISTRIBUTOR	69.2%	4,906	139	HIGH
WY	$10M	MASTER DISTRIBUTOR	69.2%	4,899	124	HIGH
AK	$12M	MASTER DISTRIBUTOR	68.2%	4,792	228	HIGH
UT	$65M	MASTER DISTRIBUTOR	68.5%	4,782	115	HIGH
MS	$52M	MASTER DISTRIBUTOR	68.1%	4,744	136	HIGH
MT	$9M	MASTER DISTRIBUTOR	67.7%	4,718	245	HIGH
NE	$19M	MASTER DISTRIBUTOR	67.7%	4,694	134	HIGH
MN	$64M	MASTER DISTRIBUTOR	65.5%	4,429	125	HIGH

What this chart shows. The top of the table is dominated by small-population states (SD, ME, IA, KS, DE, ND, AR, ID, WY, AK) with distributor HHIs above 4,500. Every row shows `MASTER DISTRIBUTOR` as the top distributor. Red shading highlights the HIGH-risk rows. This table is the "money slide" of the risk analysis — 15 states on one screen with HHI, manufacturer HHI, top-1 share, and risk flag all visible simultaneously.

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5. Phase 4: External Weather Data Ingestion

5.1 Why weather

Aramark's procurement is plausibly sensitive to weather conditions. Cold snaps push heating and chemical cleaning demand. Heavy precipitation affects disposables and sanitation. Hot months drive beverage and produce behavior. Layering weather onto the spend data lets us test whether these operational intuitions are actually visible in 12 months of procurement data, and whether weather disruption amplifies the concentration risks quantified in Phase 3.

5.2 Source — NOAA climdiv

The National Oceanic and Atmospheric Administration publishes monthly statewide climate data under the CIRS (Climatological Indices Reference Series) program, commonly called "climdiv". The files are served as plain HTTPS downloads from `https://www.ncei.noaa.gov/pub/data/cirs/climdiv/` in a fixed-width text format: one line per (state, year), with a 3-character state code, a division flag, a 4-digit year, and 12 consecutive 7-character fields for the 12 months of the year.

We used five variables from this dataset:

- `climdiv-tmaxst-*` — statewide monthly maximum temperature (°F)
- `climdiv-tminst-*` — statewide monthly minimum temperature (°F)
- `climdiv-pcpnst-*` — statewide monthly total precipitation (inches)
- `climdiv-hddcst-*` — statewide monthly heating degree days

- `climdiv-cddcst-*` — statewide monthly cooling degree days

5.3 Parsing the fixed-width files

A Python function (`parse_statewide_file`) walks each line, slices the fixed-width fields by character position, filters to 2025 and statewide-level (division flag `0`) rows, maps the 3-character NOAA state code to 2-letter abbreviations, and expands the 12 monthly values into long-format rows. Sentinel values for missing data (`-99.90`, `-9.99`, `-9999.`) are converted to `None` so they become pandas `NaN` and don't poison averages.

The final table `weather_state_monthly` has 588 rows (49 states × 12 months) and eight columns:

Column	Description
state	2-letter abbreviation
year_month	YYYYMM integer
avg_temp_f	(tmax + tmin) / 2 — monthly average temperature
total_precip_inches	monthly total precipitation
heating_degree_days	(defined below)
cooling_degree_days	(defined below)
rain_threshold	per-state 25th-percentile warm-month precipitation
snow_proxy	derived binary flag (described below)

5.4 Concept — Heating and Cooling Degree Days (HDD and CDD)

Definitions. Heating Degree Days and Cooling Degree Days are the standard climate metrics used by utilities, restaurant chains, and logistics companies to quantify cold intensity and heat intensity:

- **HDD** for a day = max(0, 65°F − daily_mean_temp). Summed over the month.
- **CDD** for a day = max(0, daily_mean_temp − 65°F). Summed over the month.

Interpretation. A cold northern state like Minnesota accumulates 1,000 to 1,500 HDD in January and February and roughly 0 in July and August. A warm southern state like Florida accumulates almost no HDD year-round but 300 to 400 CDD in summer. These variables quantify how intense a month's cold or heat actually was, which is exactly what we need to test whether demand responds to weather.

HDD and CDD are the primary cold and heat signals in our Phase 5 correlation analysis. They are not proxies — they are the peer-reviewed, industry-standard measures of the thing we care about.

5.5 The snow proxy — an honest workaround

NOAA's CIRS climdiv dataset does **not** include monthly snowfall as a statewide variable. Getting authoritative state-level monthly snowfall for 2025 would require aggregating station-level data from GHCN-Daily, which is hours of ETL work and still produces a less-reliable result than the temperature and precipitation data we already have.

Rather than going down that path, we constructed a binary derived indicator:

```
snow_proxy = 1 if (avg_temp_f ≤ 32) AND (total_precip_inches ≥ rain_threshold OR > 0
snow_proxy = 0 otherwise
```

`rain_threshold` is the 25th percentile of warm-month (>50°F) precipitation for each state, which gives us a per-state baseline for "meaningfully wet." The proxy flags months that are simultaneously cold-enough-for-snow and wet-enough-for-snow.

Validation. The 85 flagged snow-months have an average temperature of 22.6°F. The 503 non-snow-months average 58.4°F. The separation is clean, which means the proxy is at minimum a useful binary indicator of "wintry conditions."

Honest framing. `snow_proxy` is a derived cold-wet indicator, not raw snowfall data. In all correlation and regression analyses, HDD is our primary cold-intensity variable. The snow proxy is used for the binary categorical test only, and is framed transparently as a workaround rather than a direct measurement.

5.6 Coverage

`weather_state_monthly` covers 49 states. Hawaii, the District of Columbia, and Puerto Rico are not part of the NOAA climdiv statewide dataset. All Aramark spend from those locations is therefore excluded from weather joins in Phase 5.

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6. Phase 5: Weather Correlation Analysis

Phase 5 tests whether weather conditions visibly move the Aramark procurement mix, and whether concentration risk amplifies weather-driven operational stress. This section introduces one more critical concept (Pearson correlation) and one bit of notation ("pp" for percentage points).

6.1 The base dataframe — `weather_spend`

Phase 5 starts with a JOIN query combining the Phase 2 `agg_category_state_month` view (spend aggregated at the state × month × category grain) with `weather_state_monthly` on (state, year_month). The result is a 7,632-row dataframe where every row is one (state, month, category) combination with both the spend proxy and the full weather vector attached.

6.2 Concept — normalized spend

To compare spend patterns across states of very different sizes, we introduce `normalized_spend`:

```
normalized_spend = row_total_spend / sum(total_spend) grouped by (state, year_month)
```

This converts each category's absolute spend into its **share of the state-month total**. FOOD in Rhode Island in February is no longer a small number compared to FOOD in California in February — both are expressed as "share of their state's February spend."

This is essential for meaningful cross-state comparison. If we used raw dollars, California would always

dominate and no category or state pattern would be visible.

6.3 Notation — percentage points ("pp")

When we say FOOD's share rises by "2.44 pp" during snow months, we mean 2.44 **percentage points**, not a 2.44% relative increase. Concretely: FOOD's share goes from 60.69% to 63.13%. The difference is `63.13 - 60.69 = 2.44 percentage points`. A percentage point is an absolute arithmetic difference between two percentages. Using "pp" rather than "%" for such differences avoids the common ambiguity of whether a "2.44% increase" means absolute or relative.

6.4 The binary snow hypothesis

The first test splits the 588 state-months into 85 snow-proxy months and 503 non-snow months, computes the mean `normalized_spend` per category in each group, and looks at the difference.

Category Spend Share — Snow vs Non-Snow Months (normalized)

Category	Non-snow %	Snow %	Δ (pp)
FOOD	60.69%	63.13%	+2.44
MAINTENANCE AND ENGINEERING	6.44%	6.88%	+0.44
DISPOSABLES	7.02%	7.03%	+0.02
POOL EQUIPMENT AND SUPPLIES	0.08%	0.05%	-0.03
MEDICAL	0.64%	0.56%	-0.08
FURNITURE FIXTURES AND EQUIPMENT	1.25%	1.15%	-0.10
RETAIL AND PROMOTIONAL	0.32%	0.16%	-0.16
CLOTHING AND FOOTWEAR	1.27%	0.99%	-0.28
FOOD SERVICE EQUIPMENT AND SUPPLIES	2.50%	2.20%	-0.30
ROOM AND SPA	4.05%	3.72%	-0.33
CHEMICALS AND CLEANING	5.16%	4.80%	-0.36
GOLF EQUIPMENT AND SUPPLIES	0.87%	0.31%	-0.55
BEVERAGE	9.74%	9.03%	-0.71

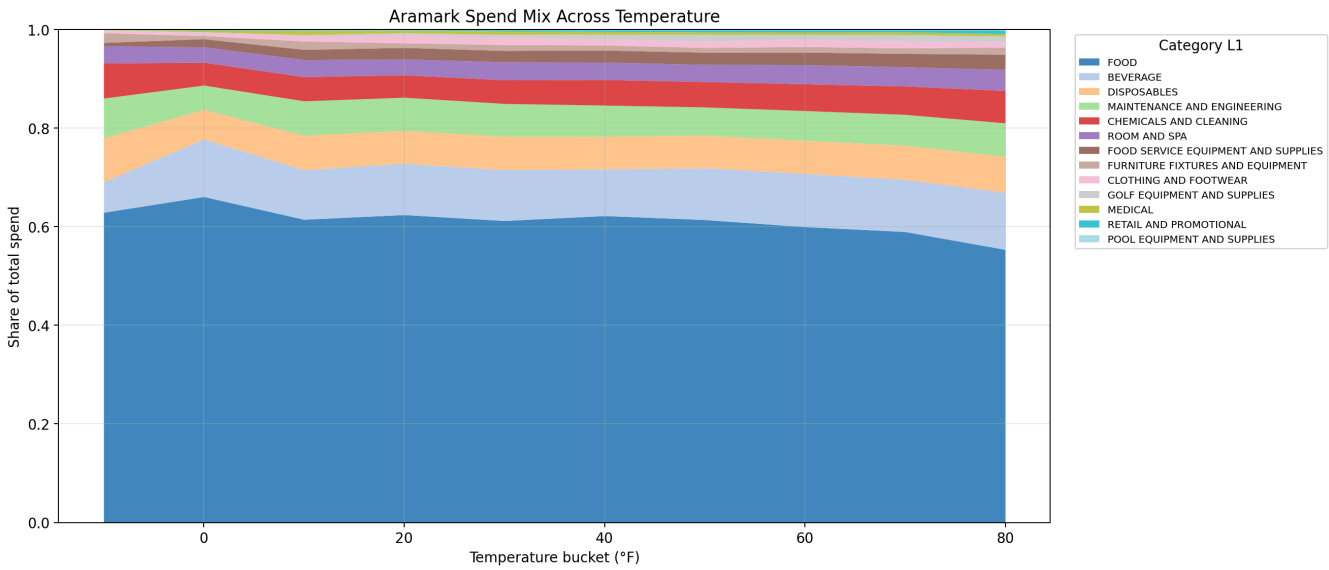
What this table shows. FOOD rises by 2.44 pp in snow months (from 60.69% to 63.13%). MAINTENANCE AND ENGINEERING rises by 0.44 pp, consistent with operational demands during winter. Discretionary categories all fall: ROOM AND SPA (−0.33 pp), FOOD SERVICE EQUIPMENT (−0.30 pp), CLOTHING AND FOOTWEAR (−0.28 pp), RETAIL AND PROMOTIONAL (−0.16 pp), FURNITURE (−0.10 pp).

The pattern is consistent with operational intuition: essentials crowd out discretionary spending during bad weather. The magnitudes are small in percentage-point terms but real given the 588-row sample. This is the entry point to the weather story.

6.5 The category mix across temperature

The binary test treats every snow month identically. A continuous version is stronger: does the mix shift smoothly with temperature, or does it jump only at the cold-wet extreme?

We bin `avg_temp_f` into 10°F buckets and plot the stacked area of category shares across temperature.



What this chart shows. FOOD's slab visibly thickens at low temperatures and thins at high temperatures. The FOOD share is roughly 69% at 0°F and drops to 54% at 80°F — a 15-percentage-point swing across the full range. BEVERAGE's slab thickens slightly as temperature rises (about 6% to 10%). DISPOSABLES stays roughly flat. CHEMICALS AND CLEANING and MAINTENANCE show subtle shifts.

This visual confirms the binary snow finding as part of a smooth continuous trend, not an artifact of the cold-wet threshold.

6.6 Concept — the Pearson correlation coefficient (r)

What it is. The Pearson correlation coefficient `r` measures the linear association between two variables. The formula:

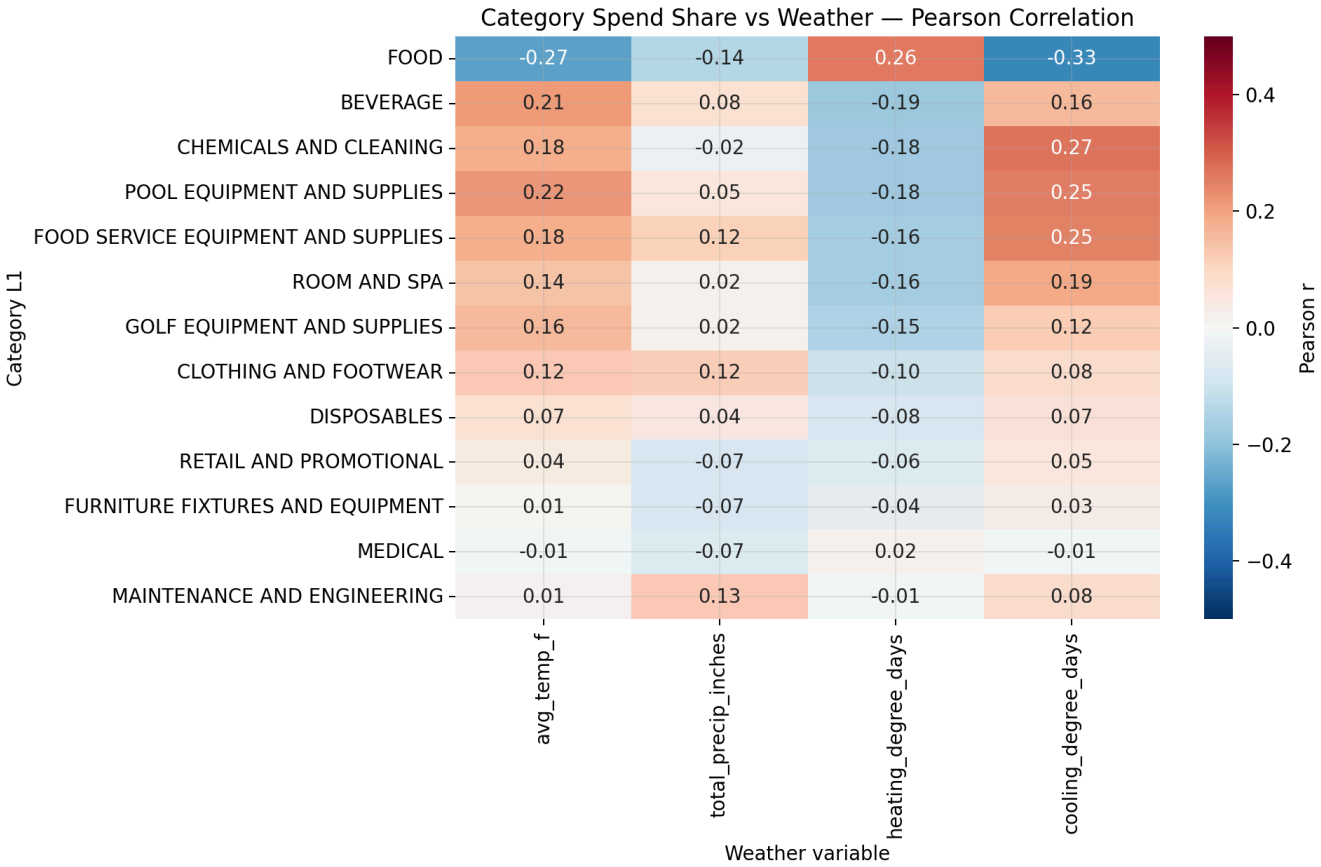
$$r = \Sigma[(x - \bar{x})(y - \bar{y})] / \sqrt{(\Sigma(x - \bar{x})^2 \times \Sigma(y - \bar{y})^2)}$$

Intuitively, `r` is a standardized version of the covariance. Values run from −1 (perfect inverse) through 0 (no linear association) to +1 (perfect direct).

Rules of thumb:

- $|r| < 0.1$ — essentially no linear signal
- 0.1 – 0.3 — weak but real
- 0.3 – 0.5 — moderate
- above 0.5 — strong

For each of the 13 product categories we compute `r` between `normalized_spend` and each of four weather variables: average temperature, total precipitation, heating degree days (HDD), and cooling degree days (CDD). With 588 (state, month) observations per category, even modest `r` values are meaningful.



What this heatmap shows. FOOD is the most weather-sensitive category in the dataset. Its correlations with the weather variables are:

- avg_temp_f: −0.27 (moderate — spend share rises as temperature falls)
- total_precip: −0.14 (weak)
- heating_degree_days: +0.26 (moderate — spend share rises as cold intensifies)
- cooling_degree_days: −0.33 (moderate, the single largest absolute correlation in the entire matrix)

Three other categories show notable heat sensitivity (positive CDD correlation):

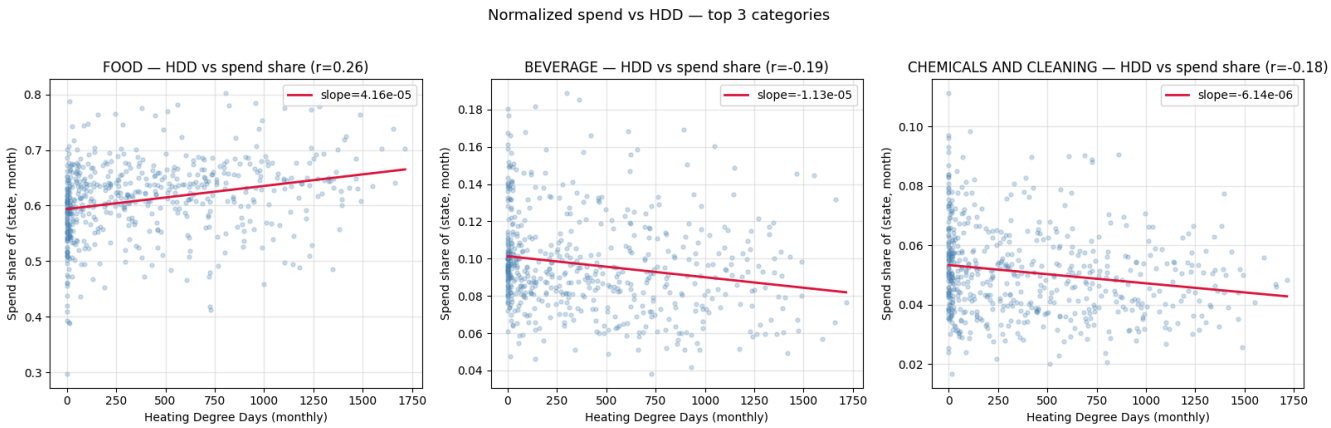
- CHEMICALS AND CLEANING: +0.27
- POOL EQUIPMENT AND SUPPLIES: +0.25
- FOOD SERVICE EQUIPMENT AND SUPPLIES: +0.25

BEVERAGE shows a weaker heat signal than operational intuition might predict: `r = +0.16` with CDD. Real, but less than FOOD's inverse heat signal.

6.7 HDD regression — cold intensity as a continuous driver

The binary snow test and the heatmap both point to cold-weather sensitivity in FOOD. A scatter plot with a linear regression line closes the loop — we show FOOD, BEVERAGE, and CHEMICALS AND CLEANING vs HDD.

A **linear regression** fits a straight line `y = slope × x + intercept` to the data by minimizing the sum of squared vertical distances between the line and the points. The slope tells you the direction and strength of the linear relationship. We compute it with `np.polyfit(x, y, 1)`.



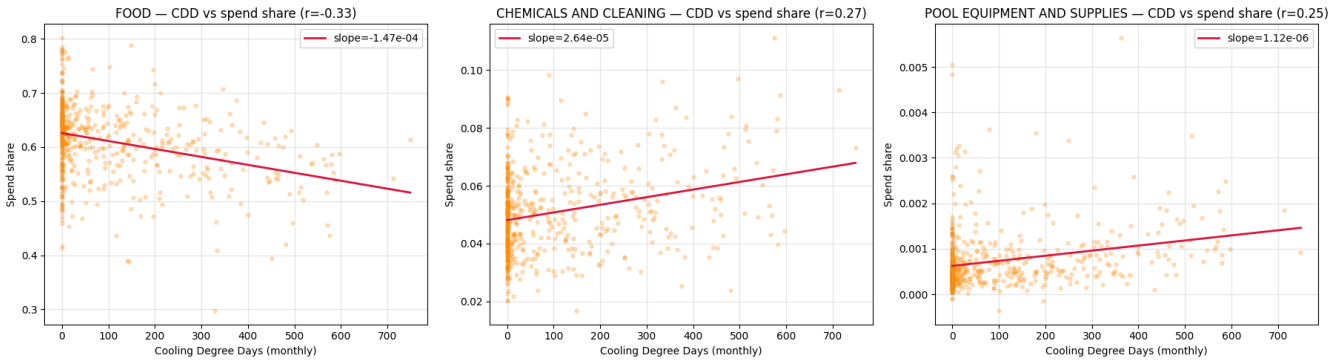
What these scatter panels show. Each dot is one (state, month) observation. The red line is the least-squares linear fit; the `r` value in the title is the Pearson correlation.

- **FOOD**: clear positive slope. Points at low HDD (warm months) cluster around spend-share 0.55–0.60; points at high HDD (cold months, especially in northern states) cluster around 0.65–0.75. `r = +0.26`.
- **BEVERAGE**: clear negative slope. Cold months compress beverage share. `r ≈ -0.19`.
- **CHEMICALS AND CLEANING**: weak negative slope. Mostly flat across the HDD range. `r ≈ -0.18`.

The FOOD regression confirms the continuous version of the cold-weather finding. It is not an artifact of the binary cutoff.

6.8 CDD regression — heat intensity and warm-weather categories

The three categories the heatmap flagged as most heat-sensitive get their own scatter panel.



What these panels show.

- **FOOD** (`r = -0.33`): the strongest single weather signal in the project. FOOD's share declines as summer heat intensifies. This mirrors the HDD finding — FOOD dominates cold months and recedes in hot months.
- **CHEMICALS AND CLEANING** (`r = +0.27`): the counterintuitive result. Cleaning chemical demand rises in summer. Plausible explanation: heavy hospitality usage, more frequent room turnover, and heat-accelerated bacterial / sanitation cycles all drive cleaning in summer.
- **POOL EQUIPMENT AND SUPPLIES** (`r = +0.25`): the sanity check. Pool equipment genuinely does spike in hot months. The fact that the methodology surfaces this validates the approach.

6.9 The risk overlay — Phase 3 meets Phase 5

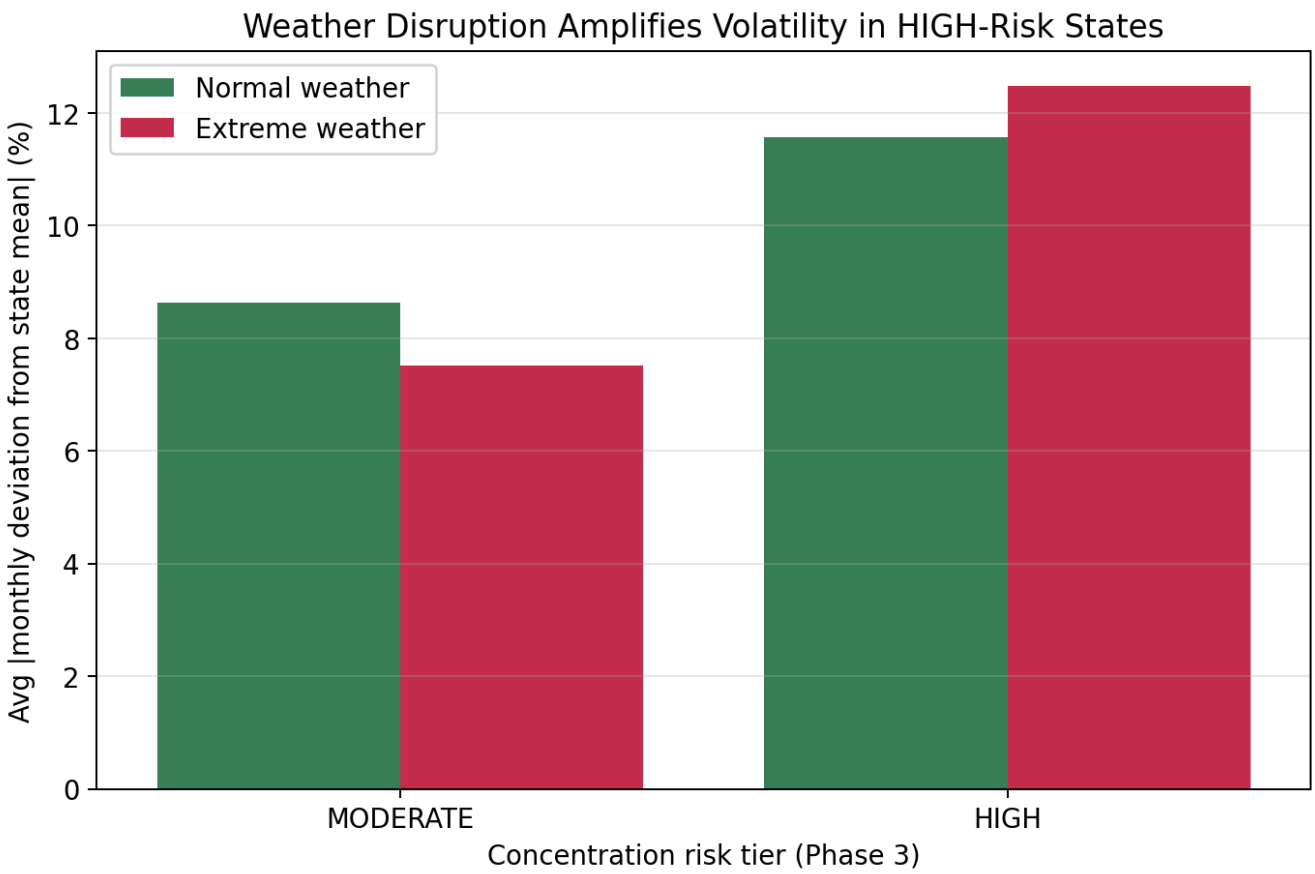
This is the integration beat. Phase 3 flagged 42 states HIGH-risk on distributor concentration. The theory is that those states have less operational flexibility during disruption — fewer alternative channels, higher dependence on `MASTER DISTRIBUTOR`. If the theory holds, HIGH-risk states should show bigger spend swings during extreme-weather months than MODERATE-risk states do.

Methodology.

1. Compute each state's total monthly spend (all categories combined).
2. Compute each state-month's absolute percentage deviation from the state's own 12-month average.
3. Flag months as "extreme weather" when HDD or CDD is in the top national quartile.
4. Split by `risk_flag` from the Phase 3 scorecard and compare mean absolute deviation across the (risk × extreme) 2×2.

Results:

Risk tier	Normal weather	Extreme weather	Δ
HIGH	11.57%	12.47%	+0.90 pp
MODERATE	8.62%	7.51%	−1.11 pp



What this chart shows. Two real phenomena simultaneously:

1. **HIGH-risk states are structurally more volatile.** Even in normal weather they swing by about 3 percentage points more month-to-month than MODERATE-risk states do.
2. **Extreme weather widens the gap.** HIGH-risk states' deviation rises during extreme weather (+0.90 pp) while MODERATE-risk states' actually falls (−1.11 pp). The combined spread is roughly 2 pp.

This is the first empirical link between the concentration risk quantified in Phase 3 and an observable operational consequence in the data. States with fewer alternative supplier channels have less operational flexibility when conditions force a change, and that vulnerability shows up in the spend numbers.

6.10 Summary of Phase 5 findings

1. **FOOD is the most weather-sensitive category.** Roughly 15 percentage-point swing in spend share across the temperature range. Strongest correlation (`r = -0.33` with CDD) in the weather matrix.
2. **Three categories rise with summer heat.** CHEMICALS AND CLEANING, POOL EQUIPMENT AND SUPPLIES, FOOD SERVICE EQUIPMENT. Of those, chemicals is the most business-meaningful.
3. **Concentration risk amplifies weather shock.** HIGH-risk states respond to extreme weather with larger spend swings. First empirical Phase 3 × Phase 5 integration.

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7. Phase 7: Machine Learning Segmentation

Phase 7 applies unsupervised clustering to discover customer archetypes hidden in the transaction data. This

section introduces several machine learning concepts in the order we need them.

7.1 Concept — clustering and K-Means

What clustering is. Clustering is an unsupervised learning method. Given a set of records, each described by a numerical feature vector, clustering partitions the records into groups (clusters) such that records within the same cluster are more similar to each other than to records in other clusters. "Similar" means close to each other in the multi-dimensional feature space.

What K-Means is. K-Means is the most widely-used clustering algorithm. It requires you to specify k, the number of clusters. It then alternates between two steps until the assignments stop changing:

1. Assign each point to the nearest cluster center (centroid).
2. Move each cluster center to the mean position of its assigned points.

K-Means minimizes **inertia**, the sum of squared distances between each point and its assigned centroid. Smaller inertia means tighter clusters.

7.2 Concept — MiniBatch K-Means and why we chose it

Standard K-Means works well up to a few hundred thousand rows but is too memory-intensive at 43 million. **MiniBatch K-Means** is the standard scalable variant: at each iteration it updates the centroids using a small random batch of points rather than the full dataset. Quality is very close to full K-Means at a fraction of the compute cost, and it scales to tens of millions of rows comfortably.

Even MiniBatch K-Means is overkill for 43M rows when the goal is discovery, not per-row assignment. A 500,000-row **stratified sample** (`RAND() < 0.012`) gives K-Means enough diversity to find clusters while fitting in memory and running in under two minutes.

7.3 Concept — feature engineering

What feature engineering is. Machine learning models don't take raw data; they take numerical feature vectors. Feature engineering is the process of turning the raw columns into the numeric representation the model needs. For categorical columns there are three common encoding choices:

- **One-hot encoding.** Replace a categorical column with N binary columns, one per category. Each row gets a 1 in the column for its category and 0s elsewhere. Preserves all information but increases dimensionality.
- **Frequency encoding.** Replace a categorical column with a single numeric column containing the fraction of rows having that category. Compact, but loses within-category differentiation.
- **Target encoding.** Replace each category with the mean target value for that category. Powerful but requires a target, which we don't have in unsupervised learning.

What the initial draft got wrong. The first version of the Phase 7 notebook used frequency encoding for `category_level_1` and `distributor_group`. That collapsed FOOD (68% of rows) and `MASTER DISTRIBUTOR` (66% of rows) into near-constants for most transactions, destroying the differentiation signal inside those fields. The resulting clusters were dominated by the one-hot `room_band` columns, producing segments that re-expressed room-size binning with identical top categories and top distributors across clusters.

The rebuild. We replaced frequency encoding with **top-5 one-hot encoding**:

- For `category_level_1`, the top-5 categories by total spend (FOOD, BEVERAGE, CHEMICALS AND CLEANING, DISPOSABLES, MAINTENANCE AND ENGINEERING) each get their own one-hot column; everything else goes into an "OTHER" bucket.
- Same pattern for `distributor_group`.
- `business_entity_type` and `room_band` are one-hot encoded straightforwardly (2 and 5 columns respectively).

This preserves within-category differentiation without exploding dimensionality. Total categorical columns: 6 + 6 + 2 + 5 = 19.

7.4 State-level enrichment — bringing Phases 3 and 4 into the model

Each transaction is joined to its state's Phase 3 risk metrics and Phase 4 weather metrics:

- From Phase 3 risk scorecard: `dist_hhi`, `top1_share_pct`, `mfr_hhi`, `risk_high_flag` (binary)
- From Phase 4 weather aggregates: `total_hdd`, `total_cdd`, `temp_volatility` (annual totals or standard deviation across the 12 months per state)

This adds 8 more numeric columns. Clusters can now differentiate along concentration exposure, climate profile, and weather variability — not just spend mix and operational scale.

Three states present in the scorecard (HI, DC, PR) are not in the NOAA weather dataset. Their transactions receive the median weather values for the missing columns so they do not break the scaling step.

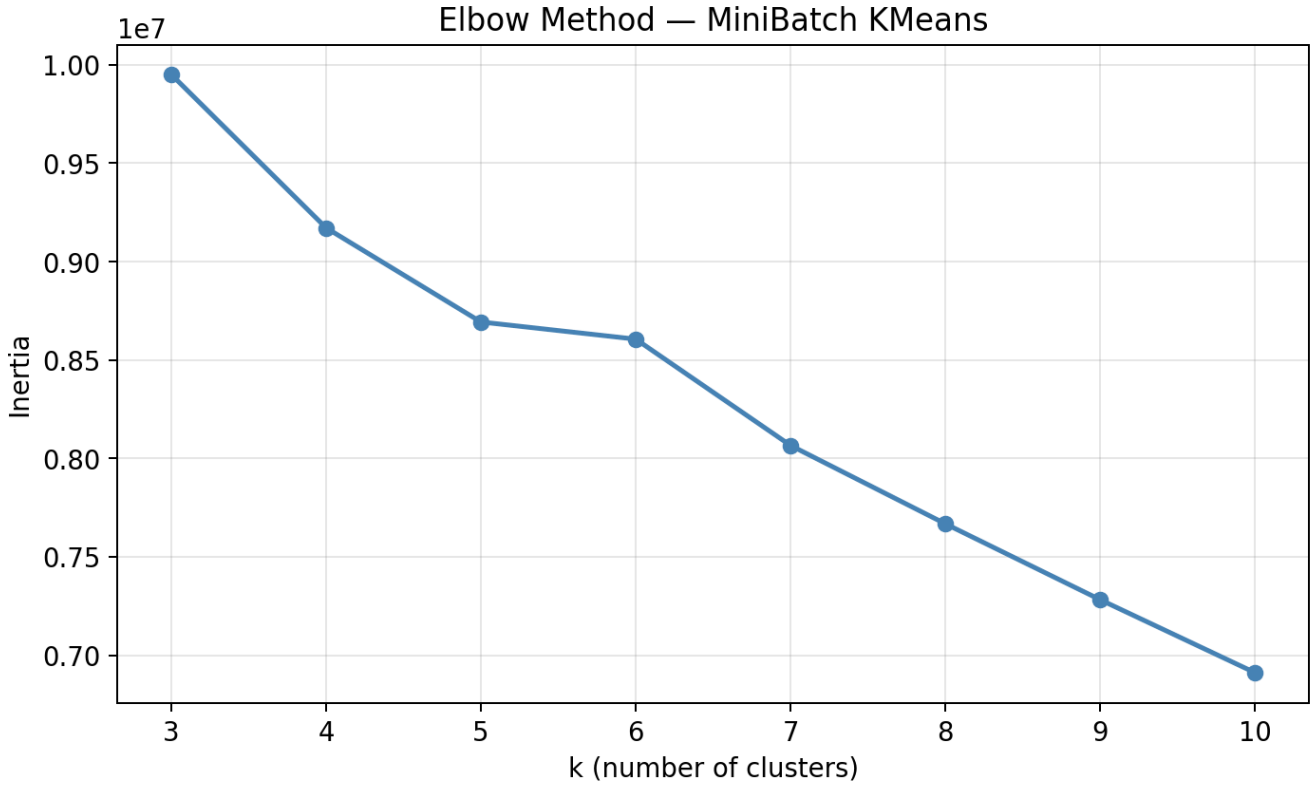
7.5 Concept — StandardScaler

What it does. K-Means is a distance-based algorithm. Without scaling, a raw HDD value in the thousands would dominate the distance calculation, drowning out 0/1 one-hot signals entirely. `StandardScaler` from scikit-learn normalizes each column to have mean 0 and standard deviation 1 — equivalent to subtracting the mean and dividing by the standard deviation.

After scaling, every feature contributes to distance on a comparable scale. This is a mandatory preprocessing step for K-Means whenever feature magnitudes vary widely.

7.6 Concept — the elbow method

What it is. K-Means requires you to specify k, the number of clusters, up front. The elbow method helps pick a defensible k. We fit K-Means models for a range of k values and plot inertia on the y-axis. Inertia always decreases as k grows (more clusters = tighter fits, trivially). The **elbow** is the point where the drop in inertia starts to diminish — adding more clusters after the elbow gives diminishing returns.



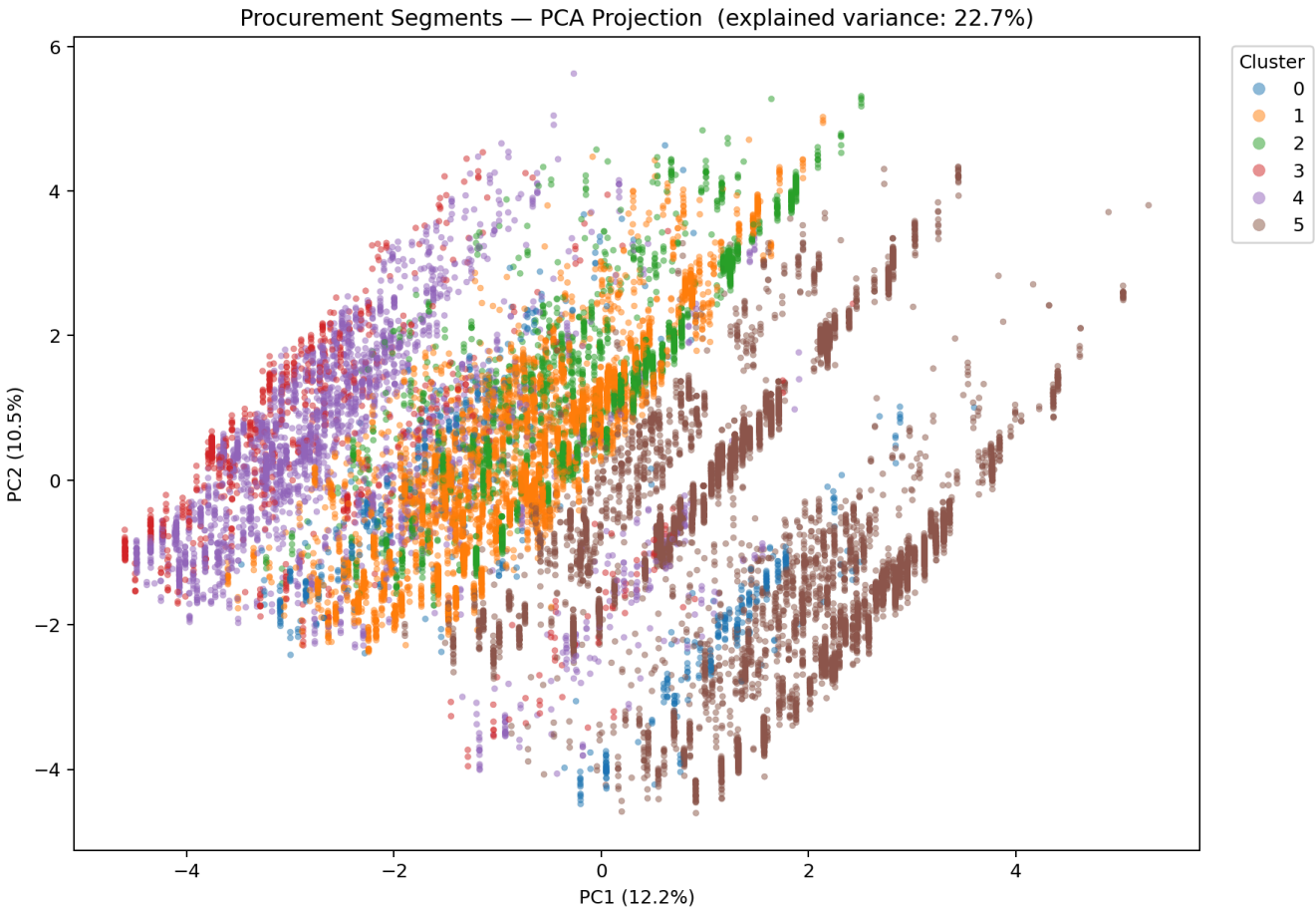
Reading our elbow chart. The inertia drop between k=3 and k=5 is steep, then flattens somewhat. There is a visible bend around k=5 to k=7, after which the curve flattens more gradually. We chose **k = 6** as a compromise between separation quality and interpretability: enough clusters to capture multiple archetypes, not so many that naming them becomes impossible in a 15-minute presentation.

A side note. The initial broken version of the clustering pipeline (with frequency-encoded features) produced a much sharper elbow at k=6. That "cleaner" curve was partly a symptom of a weaker feature space — when most of the signal lives in a small number of features (room bands), K-Means finds the natural groupings quickly and inertia flattens hard. Our richer feature space has multiple genuine axes of variation, so the curve descends more gradually. Counterintuitively, a messier elbow often indicates a better feature space, not a worse one.

7.7 Concept — Principal Component Analysis (PCA)

What PCA is. Principal Component Analysis is a linear dimensionality reduction technique. Given a high-dimensional feature space (about 27 columns in our case), PCA finds orthogonal directions (principal components) that capture the maximum variance, in descending order of how much variance each one explains. Projecting the data onto the first two principal components gives a 2D scatter plot that approximates the geometry of the full space.

We use PCA here purely for **visualization**, not for clustering. The clusters are fit on the full scaled feature matrix. PCA just gives us a 2D picture of whether the clusters actually separate in the high-dimensional space.



Reading the PCA scatter. Each dot is one transaction, colored by its cluster assignment. Visible separation between cluster regions indicates that the clusters are real — K-Means is not just partitioning a homogeneous blob. Our PCA scatter shows distinct colored regions with some overlap, which is exactly what a believable segmentation looks like. The explained variance ratio in the title tells us what fraction of the original variance the 2D projection captures.

7.8 Cluster profiling — the six archetypes

After fitting k = 6 MiniBatch K-Means, we aggregate each cluster's mean numeric features and mode categorical features on the **raw** feature columns (not the encoded ones). This is essential — profiling on one-hot or frequency-encoded columns gives unreadable numbers. Profiling on raw columns (category_level_1 = "FOOD", distributor_group = "MASTER DISTRIBUTOR") gives a profile table that reads in business language.

Phase 7 — Cluster Profile Table

Cluster	n	Mean spend	Mean rooms	Mean HHI	Top-1 share	Mean HDD	Mean CDD	% HIGH	Top cat	Top distributor	Top entity	Top rooms	Top state
3	19,168	$244	150	3,176	54.3%	3,718	1,628	83.0%	MAINT & ENG	ENG MRO	GPO	101_300	CA
0	14,812	$236	99	3,218	54.6%	3,866	1,641	85.0%	BEVERAGE	DIRECT BEV	GPO	0_non_hosp	TX
4	57,906	$205	143	3,198	54.5%	3,695	1,639	84.0%	ROOM & SPA	ENG MRO	GPO	101_300	CA
5	207,677	$173	0	3,335	55.7%	4,149	1,537	87.0%	FOOD	MASTER DIST	GPO	0_non_hosp	TX
1	139,866	$99	255	3,158	54.1%	3,781	1,589	84.0%	FOOD	MASTER DIST	GPO	101_300	CA
2	76,452	$54	64	3,449	56.7%	4,283	1,384	87.0%	FOOD	MASTER DIST	GPO	1_100	CA

The six segments:

Cluster 3 — Facility & Engineering Operations. (19,168 rows, mean spend $244, mean HHI 3,176, 83% HIGH-risk.) The highest-spend cluster. Top category is MAINTENANCE AND ENGINEERING, top distributor is ENGINEERING MRO — a specialty channel, not `MASTER DISTRIBUTOR`. Top room band is 101–300, top state CA. This segment is mid-size hospitality properties running active facility-maintenance programs.

Cluster 0 — Specialty Beverage Procurement. (14,812 rows, mean spend $236, 85% HIGH-risk.) The only cluster where BEVERAGE is the top category and DIRECT BEVERAGE is the top distributor. Non-hospitality (room band 0), top state TX. Small by row count, but structurally unique — these are beverage-specialty buyers that operate outside the main `MASTER DISTRIBUTOR` flow.

Cluster 4 — Hospitality Room & Spa Operations. (57,906 rows, mean spend $205, 84% HIGH-risk.) Top category is ROOM AND SPA, top distributor is ENGINEERING MRO. Mid-size hotels (101–300 rooms), top state CA. This segment sits alongside Cluster 3 in operational profile (same distributor, similar room band, similar spend level) but distinguishes itself through the room-and-spa category focus rather than engineering focus.

Cluster 5 — Institutional Food Backbone. (207,677 rows, mean spend $173, 87% HIGH-risk.) The largest cluster — roughly 40% of the sample. Top category FOOD, top distributor `MASTER DISTRIBUTOR`, non-hospitality (room band 0), top state TX. This is the core of Aramark's business: large institutional food service accounts (hospitals, corporate campuses, universities) routed through the central distributor channel. The Phase 3 concentration story hits this segment hardest because it is the biggest segment and routes most heavily through `MASTER DISTRIBUTOR`.

Cluster 1 — Mid-Size Hospitality Food Buyers. (139,866 rows, mean spend $99, 84% HIGH-risk.) Top category FOOD, top distributor `MASTER DISTRIBUTOR`, 101–300 rooms, top state CA. The standard hospitality food procurement segment — mid-size hotels ordering food through the central channel at moderate transaction sizes.

Cluster 2 — Small-Property Long Tail. (76,452 rows, mean spend $54, 87% HIGH-risk.) The lowest-spend cluster in the dataset. Top room band 1–100 rooms (the smallest), top state CA. FOOD and `MASTER DISTRIBUTOR` like most of the sample, but the small-property and low-transaction-size signature distinguishes this as the "long tail" segment — small hotels and restaurants with modest ordering patterns.

7.9 What the clusters reveal

1. **The feature engineering rebuild worked.** Unlike the initial frequency-encoded version, the rebuilt clusters differentiate on multiple real axes: specialty channels (Clusters 0 and 3), product focus

(Cluster 4's room-and-spa), operational scale (Cluster 2's small properties, Cluster 5's institutional backbone), and entity / channel structure.

2. **Two clusters identify specialty channels that national averages hide.** Cluster 0 (DIRECT BEVERAGE) and Clusters 3 and 4 (ENGINEERING MRO) show that beneath the `MASTER DISTRIBUTOR` dominance story there are legitimate specialty distribution paths that specific customer types already use.

3. **Concentration risk is not uniform across clusters.** All six clusters live in the HIGH-risk tier (83–87% high-risk rows), but the clusters that would be most disrupted by a `MASTER DISTRIBUTOR` failure are the two biggest FOOD/MASTER-DIST clusters (5 and 1) and the small-property tail (Cluster 2). The specialty-channel clusters (0, 3, 4) have implicit diversification because they rely on ENGINEERING MRO or DIRECT BEVERAGE for their primary category spend.

4. **Cluster geography and climate link back to Phase 5.** The top states per cluster split between TX (Clusters 0 and 5) and CA (Clusters 1, 2, 3, 4), and the weather signatures (annual HDD, CDD) line up with known climate profiles. This is consistent with the Phase 5 finding that weather-driven operational risk lands on different clusters differently.

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8. Headline Findings Across All Phases

These are the high-signal findings suitable for executive or classroom presentation. Every number below is defined and explained in the sections above.

1. **One logistics channel carries over half of Aramark's procurement spend.** `MASTER DISTRIBUTOR` holds 54.3% of national spend and is the #1 distributor in every US state without exception. This is the project's top-line finding.

2. **Every US state registers as concentrated under DOJ antitrust thresholds.** 42 of 52 states exceed HHI 2,500 (HIGH concentration tier); the remaining 10 are MODERATE. The most concentrated states are small-spend states with single-distributor shares above 70%.

3. **The manufacturer base is diversified, which reframes the risk.** Upstream manufacturer HHI is under 500 in nearly every state. If the distributor channel failed, the products themselves would still be available in the market. The risk is continuity of operations, not continuity of supply.

4. **FOOD is the most weather-sensitive category in the dataset.** Its share of (state, month) spend swings approximately 15 percentage points across the temperature range. Correlation with cooling degree days is `r = -0.33`, the largest in the weather-category correlation matrix.

5. **Concentration risk and weather shock interact measurably.** HIGH-concentration states show a 2 percentage point wider spend deviation during extreme-weather months than MODERATE-concentration states do. Phase 3 risk theory is visible in Phase 5 operational data.

6. **Customer base organizes into six interpretable archetypes.** Specialty channels (DIRECT BEVERAGE, ENGINEERING MRO) emerge as dominant for specific segments. The institutional food backbone is the largest segment and the most exposed to concentration risk.

9. Business Recommendations

The findings in this report are descriptive — they say what is true about the Aramark procurement data. This section is prescriptive. Each recommendation ties directly to a finding above and suggests a measurable action and a key performance indicator (KPI) the procurement team could use to track progress.

9.1 Pilot supplier diversification in the small-state HIGH-risk tier

Evidence. Phase 3 identified five states (SD, ME, IA, KS, DE) where the single-distributor share exceeds 70% and distributor HHI is above 5,000. These are the most operationally fragile states in the network.

Recommendation. Run a 12-month pilot in two or three of these states that intentionally routes 10–15% of category spend through an alternative distributor (an existing specialty channel or a new regional partner). Measure the cost difference and the operational resilience gain (fewer stockouts, smaller month-to-month deviation).

KPI. State-level `top1_share_pct` below 65% within 12 months, measured against the Phase 3 scorecard as a baseline.

9.2 Treat "alternative channel readiness" as a tracked operational metric

Evidence. Every US state in the dataset exceeds DOJ's moderate concentration threshold. 42 of 52 are HIGH risk. This is a systemic pattern, not a localized issue.

Recommendation. Establish regional backup-distributor agreements with clear activation procedures. Track the time it would take to shift a non-trivial share of spend (say, 20%) to an alternative channel in each region. Make this a line item in monthly procurement reviews rather than a disaster-recovery plan that sits unopened.

KPI. Activation time (estimated days to reroute 20% of spend) per region, tracked quarterly.

9.3 Focus risk management investment on logistics redundancy, not re-sourcing

Evidence. Phase 3's manufacturer HHI analysis showed upstream concentration is low — manufacturer HHI is below 500 in nearly every state. The products Aramark buys are broadly available in the market. The concentration bottleneck sits in the distribution layer only.

Recommendation. Resist the temptation to respond to concentration risk with supplier-contract renegotiation. The products themselves are not scarce. Invest instead in logistics redundancy: secondary distribution agreements, targeted inventory buffers in high-volatility states, and logistics-capacity pre-commitments during forecasted disruption windows.

KPI. Ratio of risk-mitigation budget allocated to logistics redundancy vs. procurement contract renegotiation — target 70/30 or better.

9.4 Integrate weather forecasts into monthly replenishment cadence

Evidence. Phase 5 quantified FOOD's weather sensitivity at `r = -0.33` with cooling degree days — the strongest single correlation in the matrix. FOOD's share of state-month spend swings approximately 15 percentage points across the temperature range. CHEMICALS AND CLEANING, POOL EQUIPMENT, and

FOOD SERVICE EQUIPMENT all rise with summer heat.

Recommendation. Build a weather-responsive replenishment trigger using NOAA's monthly HDD/CDD forecasts. Increase FOOD inventory in pre-HDD months (October and November in northern states). Increase CHEMICALS, POOL, and FOOD SERVICE EQUIPMENT inventory ahead of forecasted high-CDD months in warm states. This is a moderate operational change that can be piloted in a single state before rolling out.

KPI. Stockout incidents during the first seven days of weather-anomaly months. Measure against the prior-year baseline from the same state.

9.5 Activate extreme-weather contingency procedures proactively

Evidence. The Phase 5 risk overlay showed that HIGH-concentration states register a 2 percentage point wider spend deviation during extreme-weather months than MODERATE-concentration states do. The gap is small in absolute terms but is the signal of avoidable operational stress — HIGH-risk states have less flexibility to absorb shocks.

Recommendation. Tie the activation of contingency distribution procedures to forecasted (not observed) extreme-weather conditions. When NOAA forecasts a top-quartile HDD or CDD month for a HIGH-risk state, open pre-arranged alternative channels before the month begins rather than after a disruption occurs. The cost of activation is small; the cost of reactive response is the 2 pp spend deviation we measured.

KPI. Percentage of forecasted extreme-weather months where contingency procedures were activated proactively. Target 100% within two years.

9.6 Study what makes the specialty-channel clusters successful and transfer patterns to mass-market segments

Evidence. Phase 7 surfaced two clusters (Cluster 0 — Specialty Beverage Procurement, and Cluster 3 — Facility & Engineering Operations) that already operate outside the dominant `MASTER DISTRIBUTOR` channel. They use DIRECT BEVERAGE and ENGINEERING MRO respectively for their primary category spend. These customers have implicit diversification that the mass-market FOOD clusters (5 and 1) lack.

Recommendation. Commission a qualitative study of Cluster 0 and Cluster 3 customers: how did they end up on specialty channels, what is their cost and service experience, and what barriers prevent the same pattern in mass-market FOOD procurement? Use the findings to design a targeted migration offer for a subset of Cluster 1 (Mid-Size Hospitality Food Buyers) customers.

KPI. Share of Cluster 1 customers with at least 10% of category spend on a non-`MASTER DISTRIBUTOR` channel within 18 months.

9.7 Maintain an honest data caveat log alongside reporting

Evidence. Several limitations constrain the analysis: only 12 months of data, a randomized spend proxy, three states missing from the weather join (HI, DC, PR), 34% missing manufacturer IDs, and non-reproducible BigQuery `RAND()` sampling in the ML pipeline.

Recommendation. Rather than hiding these limitations, publish them alongside any downstream business recommendation that depends on them. Industry reviewers will trust the work more, not less, when caveats are visible. The honest framing is that this analysis describes one calendar year of directional patterns — a strong

foundation, but not a substitute for multi-year validation.

KPI. Every downstream business memo or KPI proposal derived from this analysis includes a cited limitations section — measured by audit, not self-report.

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10. Executive Summary

This is the high-level recap, suitable for reading after all the concepts above have been introduced. Every term here is defined in the preceding sections.

The project analyzes roughly 43 million rows of 2025 Aramark US procurement data, organized into seven phases that build a cloud-native analytics pipeline (Google Cloud Storage → BigQuery warehouse → 16 aggregation views → Colab notebooks for concentration analysis, weather correlation, and ML segmentation).

The headline findings:

1. **Supply channel concentration is extreme.** The top distributor group (`MASTER DISTRIBUTOR`) handles 54.3% of all Aramark spend nationally and is the number-one distributor in every single US state. The top 10 distributor groups together account for 86.9% of total spend. The Gini coefficient on distributor spend is well above 0.8 — the "extreme concentration" band.
2. **Every US state falls into a DOJ-defined concentration risk tier.** 42 of 52 states are flagged HIGH risk (distributor Herfindahl-Hirschman Index above 2,500) and the remaining 10 are MODERATE. Zero states are classified as competitive. The most concentrated states are small-spend states (SD, ME, IA, KS, DE) with single-distributor shares above 70%.
3. **Manufacturer diversity is the reassuring finding.** Upstream of the distributor layer, manufacturer concentration is low across nearly all states (HHI typically under 250). If `MASTER DISTRIBUTOR` were disrupted, the products themselves would still be available from multiple manufacturers in the market. The risk is continuity of operations, not continuity of supply.
4. **Weather measurably shifts the procurement mix.** FOOD's share of (state, month) spend rises approximately 2.44 percentage points during cold-wet months (binary snow test) and correlates negatively with cooling degree days at `r = -0.33` — the strongest single correlation in the weather-category matrix. Specialty categories like chemicals and pool equipment rise with summer heat ($r \approx$ +0.25 to +0.27).
5. **Concentration risk and weather shock interact.** HIGH-concentration states show a 2 percentage point larger absolute spend deviation during extreme-weather months than MODERATE-concentration states do. This is the first empirical link between the theoretical risk quantified in Phase 3 and observable operational consequences in the data.
6. **Customer segmentation reveals six distinct procurement archetypes.** MiniBatch K-Means clustering on an enriched feature space produces segments that differentiate along spend intensity, concentration exposure, distributor channel, operational scale, and weather context. Specialty distribution channels that are invisible in the national average (DIRECT BEVERAGE, ENGINEERING MRO) emerge as dominant in specific customer segments.

The combination of these findings gives a coherent business narrative: Aramark's procurement runs through a

highly concentrated logistics channel in every US state, that concentration amplifies volatility during weather disruption, and specific customer archetypes are more or less exposed to this risk depending on which specialty channels they use. The manufacturer layer itself is diversified, which reframes the risk as operational rather than supply-sourcing.